Invest now, get paid later? Limits and opportunities of woody plants to time growth in a future climate

Frederik Baumgarten[1-3], Sally Aitken[1], Yann Vitasse[3], Robert D Guy[1], EM Wolkovich[1]

September 22, 2025

[1] Department of Forest and Conservation, Faculty of Forestry, University of British Columbia, 2424 Main Mall Vancouver, BC Canada V6T 1Z4.

[2] Department of Environmental Sciences-Botany, University of Basel, Schönbeinstrasse 6, Basel 4056, Switzerland.

[3] Swiss Federal Institute for Forest, Snow and Landscape Research WSL, Zürcherstrasse. 111, Birmensdorf 8903, Switzerland

Corresponding Author: Frederik Baumgarten; frederik.baumgarten@unibas.ch;

Tel.: +41 61 207 23 37

E-Mail-Addresses: sally.aitken@ubc.ca; yann.vitasse@wsl.ch; rob.guy@ubc.ca; e.wolkovich@ubc.ca

Type of article: Perspective

Short running title: (In)determinate growth strategies of trees

Keywords: plant growth, tree phenology, shoot extension, indeterminate growers, carbon sequestration, growing season length, drought, genetic programming, phenotypic plasticity, forest productivity

wordnumbers: abstract: 200; main text: 4005; box1: 404; number of figures: 4; number of tables: 0

number of references: 84

Authors contribution: FB and EM initiated this project. FB took a lead in drafting, designed figures and performed phylogenetic analyses. All authors contributed substantially to the writing of the article.

Data accessibility statement: The data supporting the phylogeny are archived in the following public repository. Link XXX

Conflict of interest statement: The authors declare no conflict of interest

Abstract

When and how much organisms grow given environmental constraints are fundamental questions in biology, and especially pressing in the context of climate change as we strive to predict biomass production and carbon sequestration. While environmental factors such as temperature and water availability directly regulate plant growth, the timing and rate of growth are also governed by a plant's phenological sequence—its genetically programmed developmental stages. This sequence is critical for predicting climate responses but is often underappreciated.

Here, we leverage the concept of (in-)determinacy in growth and development—which captures how flexibly plants produce organs from preformed cells versus initiating them *de novo*—to propose a new framework for predicting tree growth responses to climate change.

We hypothesize that: 1) determinate growth is an adaptation to strongly seasonal climates, restricting tissue investment to a short favorable period before dormancy; 2) indeterminate species are more exposed to increasing climatic extremes (spring/fall frost, drought) but may recover more effectively from resulting damage; and 3) species with greater indeterminacy are more likely to benefit from lengthening frost-free growing seasons. Thus, future carbon sequestration may depend not only on environmental changes, but also on the degree of determinacy set by intrinsic genetic programming.

Introduction

Investing the right amount of resources in growth at the right time is of crucial importance to the survival and fitness of any living organism. The topic of growth strategies and habits has a long history in science, spanning the fields of genetics, genomics, physiology and ecology across the animal and plant kingdoms (Stearns, 1998). At its core lies the concept of determinacy—the classification of organisms as either reaching a fixed size at maturity or continuing to grow throughout their lives. Like mollusks, fish and reptiles, plants add to their primary bodies as long as they live and are therefore considered 'indeterminate growers' (Ejsmond *et al.*, 2010).

Various terms have emerged to describe this phenomenon across spatial and temporal scales, e.g. from a single cell to the whole organism, and from a single season to an entire lifetime (McDaniel, 1992; Karkach, 2006). In plants, organs can develop from preformed embryonic tissue (e.g. primordia in overwintering buds), form *de novo*, or both, depending on species and life form. While these fundamental physiological constraints apply broadly across the plant kingdom, they play a particularly central role in the life cycle of woody perennials, especially trees, which must balance growth, survival, and reproduction over long lifespans in highly seasonal environments. Yet, it remains unclear which point along this trait continuum—from strictly determinate to highly indeterminate growth—will prove more advantageous in a future climate characterized by increased environmental stress and prolonged growing seasons. To provide context, we briefly introduce the fundamental limitations to plant growth in extra-tropical regions by linking external environmental drivers and internal growth control mechanisms.

Seasonality of temperature, soil moisture and light

The further one travels from the equator towards the poles, the more strongly plants are confined to a shrinking time window of opportunity set primarily by low temperatures. Below c. 5 °C, metabolic activity slows to an extent where growth and development largely cease (Schenker *et al.*, 2014; Rossi *et al.*, 2008; Körner, 2008). More importantly, sub-freezing temperatures can cause severe damage to plant tissue if they occur at vulnerable developmental stages, e.g. during or after leaf unfolding, prior to fruit maturation, or before cold acclimation in fall (Sakai & Larcher, 1987; Baumgarten *et al.*, 2023). While annual plant species accommodate their entire life cycle within this window, perennial (woody) species are forced to partition their growing phase seasonally, with periods of activity alternating with periods of dormancy (Delpierre *et al.*, 2016). This is referred to as intermittent or rhythmic (as opposed to continuous) growth.

During the active growing season, high temperatures can also directly inhibit growth and development, by surpassing species-specific physiological thresholds (O'Sullivan *et al.*, 2017). Indirectly, high temperatures often reduce soil moisture, leading to water stress that can slow or stall growth (Hsiao, 1973; Pugnaire *et al.*, 1999; Etzold *et al.*, 2021), and in extreme cases, cause leaf scorching or premature senescence (Estiarte & Peñuelas, 2015). Together, these temperature and soil moisture limitations act as environmental filters, narrowing the period during which growth is possible (Figure **??**).

Light also influences plant growth through two key processes. First, light intensity provides the primary energy source that drives photosynthesis and, consequently, source activity. Second, light carries a wealth of information through its quality and the length of daily light and dark periods (photoperiod), which mediate many physiological processes, including bud set and dormancy transitions (Wang *et al.*, 2024). Both factors may contribute to the common observation that tree growth often peaks near the summer solstice in extra-tropical regions (Rossi *et al.*, 2006; Etzold *et al.*, 2021; Luo *et al.*, 2018) or that a window of temperature sensitivity 'opens' around the summer solstice to synchronize reproductive efforts over very large spatial scales (Journé *et al.*, 2024).

Internal programming of plants

Given our relatively robust understanding of environmental influences on plant growth, it may seem straightforward to predict how plants will grow under current and future climatic conditions. Yet, this has proven challenging, particularly when predicting responses under scenarios with extended, climatically favorable growing seasons (Zohner *et al.*, 2021). Here we propose that accurate predictions require incorporating another key, but often overlooked, factor: internal growth control—the genetically encoded developmental program that may prevent further growth *despite* favorable environmental conditions.

While plants have evolved numerous morphological adaptations to tolerate or avoid harmful environmental conditions, most temperate and boreal woody perennials cope with seasonal fluctuations in temperature and moisture by temporally escaping these conditions. They do this by aligning their life-history events (phenology) with a cycle of growth and dormancy that balances survival, reproduction, and growth over the long lifespan typical of many tree species (Delpierre *et al.*, 2016). Hence, the phenological sequence can impose abrupt internal switches in resource allocation—from vegetative growth to reproduction (flowering, fruit maturation) and storage (Stearns, 1989; Chapin *et al.*, 1990). These transitions act as additional internal filters, further narrowing the effective window during which vegetative growth can occur, regardless of environmental potential (Figure **??**).

Aim of this perspective

Climate change is extending the length of the growing season while simultaneously increasing the frequency and severity of environmental stressors such as drought (Hao *et al.*, 2018) and presumably also late spring frost events in many regions worldwide (Zohner *et al.*, 2020b). How are these potential opportunities and threats linked to species-specific growth performance, particularly for woody perennials such as trees that must balance preformed versus *de novo* shoot tissue production within each growing season? Which strategy benefits most from an extended growing season, and which can cope better with increased environmental stress?

The role of primary growth has been widely neglected in addressing these questions, even though it drives crown expansion and, consequently, total leaf area and photosynthetic capacity, which together largely determine biomass production (Girard *et al.*, 2011; Soolanayakanahally *et al.*, 2015). In this perspective we revisit classical concepts of shoot growth patterns in woody perennials, generalize them across meristem types and position them within the context of climate change. We propose that mechanisms of internal growth control—especially those governing the degree of determinacy—are key features shaping how trees and other long-lived woody species will respond to future climatic conditions.

The concept of (in)determinate growth

Full stop or temporary break in (vegetative) growth

In annual plants, growth ends with the production of flowers to form fruits and seeds: a signal in the apical meristem triggers a sudden switch in resource investment from vegetative growth to building a reproductive structure with no point of return, signaling the end of the life-cycle (Poethig, 2003; Huijser & Schmid, 2011). A shoot axis that terminates in a flower or other modified structure and ceases meristematic activity is considered 'determinate' (Barthélémy & Caraglio, 2007). In woody perennial species, however, flowers are typically produced on lateral shoots or buds, preserving the main vegetative axis and allowing continued structural expansion. However, 'determinate growth' also refers to a temporal suspension of the meristem, which leads to a time lag between the formation of organs and their expansion (Kozlowski, 2012; Hallé *et al.*, 1978).

When to form (new) shoots

In their first year, all seedlings exhibit indeterminate growth. In subsequent years, many temperate and boreal tree species preform their new shoot increments during the previous growing season, overwintering them as primordial shoots with leaves and internodes in hardened buds to be 'ready-to-go' when spring arrives. Once the leaves flush and the internodes elongate, some species suppress further activity of the apical meristem for the rest of the season through internal control mechanisms such as paradormancy (Lang *et al.*, 1987), often regulated by photoperiod-induced hormonal changes (Böhlenius *et al.*, 2006). Other species, by contrast, continue to produce neoformed leaves and internodes on top of the preformed shoot, sometimes stretching their indeterminate growth well into autumn until low temperature eventually forces them to stop. Indeterminate growth can occur either through the continuous activity of the shoot apical meristems forming new leaves and internodes without buds, termed free growth, or through the formation and subsequent flushing of buds mid-season without a period of dormancy, called second flushing or lammas growth (Figure **??**).

As we use it here, (in)determinacy in trees refers to the ability to:
a) preform organs in one year as a future investment, which is deployed rapidly in the following spring with no further primary shoot growth during that season (determinate strategy)
b) maintain a continuous or episodic meristem activity by forming new shoot tissue during the current

growing season (indeterminate strategy).

Although this concept is often presented dichotomously (Kozlowski & Pallardy, 1997; Lechowicz, 1984), but see Kikuzawa, 1983; Damascos *et al.*, 2005), with species classified as either determinate or indeterminate growers, most species likely exist along a gradient with numerous intermediate forms. For instance, many oaks (*Quercus spp.*) are considered determinate growers, but can exhibit multiple flushes. Similarly, Douglas-fir (*Pseudotsuga menziesii*) gradually adopts a more determinate growth habit with maturity (Borchert, 1976; Heuret *et al.*, 2006).

Evolutionary trade-offs: the cost of certainty and flexibility

In most temperate and boreal ecosystems, determinate and indeterminate growth strategies are successful and co-occur, raising the question of their fundamental trade-offs. The enhanced growth potential of indeterminate species should allow them to easily outgrow competing species. Faster growth, however, often comes with higher turnover rates and shorter life spans (Brienen *et al.*, 2020; Millet *et al.*, 1999). Not surprisingly, indeterminate species are frequently found to be early successional pioneers (Marks, 1975; Boojh & Ramakrishnan, 1982). Moreover, indeterminate growth enables greater phenotypic plasticity, as tissues are produced in real time in response to prevailing conditions. This flexibility can be advantageous in variable or disturbed environments where rapid resource exploitation is critical. However, this extended growth period and real-time tissue production can also increase the risk of damage from late spring or early autumn frosts, as well as exposure to drought later in the growing season, potentially offsetting the benefits of flexible growth (Figure **??**).

In contrast, determinate species preform tissues in advance and are thus more constrained by past conditions, particularly those experienced during bud formation. This strategy can reduce short-term plasticity but may enhance survival in predictable seasonal climates by protecting sensitive tissues and synchronizing growth with the most favorable part in the seasons. As a result, determinate species may exhibit stronger local adaptation and reduced short-term plasticity compared to indeterminate species (Leites & Benito Garzón, 2023). For example, western larch (*Larix occidentalis*) achieves much of its primary shoot growth from indeterminate growth and is less locally adapted than the sympatric species Douglas-fir, which grows more determinately (Roskilly & Aitken, 2024).

Together, these contrasting strategies illustrate a trade-off between maximizing growth potential and maintaining local specialization, implying ecological niche separation along gradients of disturbance and climate predictability/seasonality. With ongoing climate change, clarifying how trees control their degree of (in)determinacy will be key to understanding which growth strategy may prove most successful under future conditions.

Control mechanisms of (in)determinacy

Despite a century of research into plant growth habits and strategies, we still have a limited understanding of when and why trees exhibit a particular degree of (in)determinacy. This is likely due to the high variability of environmental conditions within and across years, sites, and individuals, which complicates efforts to disentangle internal growth regulation from external environmental influences. Favorable conditions, particularly high soil moisture availability throughout the growing season, can prolong or boost shoot elongation, sometimes enabling an additional flush (Kaya *et al.*, 1994). These findings indicate that shoot growth may come to a halt because water supply cannot support expanding leaf area. An imbalance in the root:shoot ratio under a given water status of the plant can suppress shoot growth (Borchert, 1973). By the same mechanism, many species are able to produce new shoots to rebuild the canopy after a considerable loss of leaf area due to herbivory, hail storms or late spring frosts (Baumgarten *et al.*, 2023).

However, environmental conditions do not simply alter a species' growth strategy. While a certain plasticity of apical shoot growth has been shown, this flexibility is often restricted to a specific developmental time window—and is far more limited in determinate species. This reflects a genetically fixed strategy, likely evolved as a safeguard against regular environmental stress (e.g., frost, drought; Figure **??**). In contrast, indeterminate species retain a more open and responsive growth pattern, allowing them to resume or extend growth later in the season when conditions permit. Provenance trials reinforce this idea, showing that the frequency of second flushes is higher in populations from southern origin (Rudolph, 1964).

Latitudinal variation in the frequency of second flushes suggests a photoperiodic regulation comparable to that of bud set in poplar species (Soolanayakanahally *et al.*, 2013). Bud set, growth cessation, and flowering induction appear to be regulated by a shared set of genes (*CONSTANS* and *FLOWERING LOCUS T*) regulated by photoperiod (Böhlenius *et al.*, 2006). Moreover, Wang *et al.* (2024) found that plants distinguish between absolute and photosynthetically relevant photoperiods to independently regulate flowering and vegetative growth.

Although photoperiod appears to be a key regulator of seasonal growth and development in many species, most of our knowledge comes from model organisms, with poplar being the main woody perennial studied in detail. Thus, genetic control mechanisms of (in)determinacy that set the potential of tree growth across species are yet to be discovered.

The performance of (in)determinacy with climate change

Spring warming has advanced the onset of leaf emergence by up to a month compared to pre-industrial times (Vitasse *et al.*, 2022). In contrast, autumn phenology of growth and leaf senescence has not shifted as much as one might expect, given the extension of favorable growing conditions (Zani *et al.*, 2020; Zohner *et al.*, 2023). In fact, phenological sequences are increasingly observed to shift as a whole toward spring, rather than stretching at both ends of the growing season (Keenan & Richardson, 2015; Meng *et al.*, 2024), not necessarily leading to increased biomass production during longer growing seasons (Zani *et al.*, 2020).

We hypothesize that determinate species will largely maintain their fixed growth schedules under climate warming, with no or little reductions in overall productivity, due to increased drought and heat stress with little ability to compensate (Figure **??**). In contrast, indeterminate species are likely capable of extending their growth period, potentially resulting in higher productivity under future climatic conditions (Caffarra & Donnelly, 2011; Richardson *et al.*, 2009; Zohner *et al.*, 2020a, Figure **??**). Such a phenological extension seems common for early successional species, which tend to leaf out earlier in spring and senesce later in autumn than late-successional species, likely due to their generally lower chilling and forcing requirements for bud burst and dormancy release (Laube *et al.*, 2014; Hu *et al.*, 2022; Baumgarten *et al.*, 2021; Heide, 1993).

However, the greater flexibility (i.e. phenotypic plasticity) of indeterminate species could also increase their exposure to extreme climatic events. Deciduous indeterminate species are often among the first to leaf-out and among the last to shed their leaves—occasionally as a result of first freezing events in autumn. In addition, a substantial part of their growth period falls into summer months, when the risk of drought is typically highest (Figure **??**). Therefore, we hypothesize that the conservative strategy of determinate growth largely escapes unfavorable growing conditions by growing between the last spring frost and the increasing water shortages in summer, with relatively ample safety margins in contrast to indeterminate species. Consequently, productivity shows little variation from year to year and will largely remain constant in a future climate or even decline due to higher water stress during their restricted window of potential growth (Silvestro *et al.*, 2023, Figure **??**).

Once hit by an extreme event that damages leaves, e.g. following severe drought, determinate species might not recover easily. Because canopy development of determinate, late successional species is largely completed early in the season, the capacity to produce new foliage for recovery later in the same year is limited (Baumgarten *et al.*, 2023). While some species can initiate epicormic or adventitious shoots following injury, this capacity varies widely across taxa (Meier *et al.*, 2012). It remains unclear whether species exposed to recurrent defoliation, e.g. by endemic insects outbreaks, are more likely to exhibit indeterminate growth habits. Consequently, defoliation occuring in late summer may substantially reduce photosynthetic capacity during the period when reserves are typically replenished, potentially affecting future performance.

In contrast, the flexible growth schedule of indeterminately growing species may allow plants to: 1) produce tissue better adapted to harsh environmental conditions as it is formed in the current season (e.g., smaller leaves with higher specific leaf area), and 2) catch up and compensate later in the season by another productivity boost (Stevens & Lindroth, 2005). Evidence that trees can take advantage of a 'second growing season' after summer drought was found in pines in Mediterranean climates through polycyclic flushing—a form of indeterminate growth (Figure **??** Girard *et al.*, 2011).

We argue that new opportunities and challenges for trees with climate change will increasingly disrupt their phenological cycles, favoring species and genotypes that are more plastic in rearranging their activities by resuming growth and/or storage restoration (e.g., starch, nutrients) later in the year, thereby recovering from and compensating for some stress-induced damages and losses. Future climate will therefore likely intensify the competition among co-occurring species and might re-assemble forest communities increasingly composed of species adopting an indeterminate growth strategy.

(In)determinacy beyond leaf tissue

Although (in)determinate growth is typically discussed in the context of the shoot apical meristem, similar patterns may exist in secondary growth and below-ground meristems. For example, in the vascular cambium, a timely transition from earlywood to latewood production—and from vegetative growth to reproductive or storage allocation—may reflect an internal growth schedule similar to that of primary shoot growth. Cambial initial cells divide to form daughter cells, which subsequently produce cohorts of precursors that then mature into functional xylem and phloem (Valdovinos-Ayala *et al.*, 2022). Hence the number of initial cell divisions largely determines the amount of total xylem formed over the season (Lupi *et al.*, 2010). Interestingly, several studies report that the production of new xylem cells (via cambial division) ceases even though environmental conditions remain favorable and the canopy is still photosynthetically active—suggesting that internal controls may limit secondary growth regardless of external potential (Buttò *et al.*, 2020; Arend *et al.*, 2024).

Regarding below-ground meristems, roots seem to follow a much more opportunistic and indeterminate strategy with root meristems remaining active throughout the year. Experimental evidence from rhizotron studies show root growth during mid-winter whenever soil temperatures are suitable (Lyford & Wilson, 1966), suggesting that roots do not enter true dormancy (Radville *et al.*, 2016; Marchand *et al.*, 2025). However, the degree to which asynchrony between above- and below-ground meristems reflects environmental conditions (e.g., soil temperature), root:shoot imbalances, or genetically fixed strategies remains unresolved (Abramoff & Finzi, 2015; Makoto *et al.*, 2020; Silvestro *et al.*, 2025; Campioli *et al.*, 2024).

Box 1: Metrics of (In)determinacy

To advance our understanding of growth strategies under climate change, we need not only improved methods for quantifying the degree of (in)determinacy—but also more empirical data to apply and validate these approaches across species and ecosystems. Moving beyond a simple binary classification, we propose a set of metrics that span scales, from organ-level traits to landscape-scale proxies:

Ratio of preformed to neoformed organs: The ratio of the number of leaves (or leaf scars) at the end of the season to the number of leaf primordia initially formed in buds provides a direct measure of (in)determinacy. Values greater than one indicate a higher degree of indeterminate growth. First described over a century ago (Moore, 1909), this metric has been used in only a few studies (Damascos *et al.*, 2005; Kikuzawa, 1983; Guédon *et al.*, 2006). Although quite distinct and easily counted in some species (e.g. maples; Critchfield 1971), there are only slight morphological differences between neoformed and preformed leaves in others (e.g., black cottonwood). Time-lapse imaging, labeling or artificial wounding can facilitate the distinction between preformed and neoformed organs. The number of leaf primordia can be determined through bud dissection, but this precludes the counting of neoformed leaves that would later arise from the same meristem. Micro-CT imaging may offer a modern, non-destructive alternative to bud dissection (Kovaleski *et al.*, 2019).

Seasonal growth patterns: Temporal tracking of shoot elongation and wood formation—via micro-coring, dendrometers, or regular shoot measurements—reveals when and how new tissue is produced. Wood anatomical analyses distinguish cell formation from later development, especially useful in combination with artificial wounding of the cambium that acts as a 'marker in time' (pinning technique; Gärtner & Farahat 2021). High-resolution LiDAR and photogrammetry are increasingly capable of capturing shoot elongation and structural changes across whole crowns or stands (Zhang *et al.*, 2019; Jin *et al.*, 2022).

Seasonal trends in canopy greenness: Satellite and drone imagery can track vegetation green-up and senescence at large spatial scales. For example, Meng *et al.* (2024) found that time allocation between green-up and senescence remained consistent across temperate ecosystems, despite substantial variation in growing season length and warming intensity. Such temporal dynamics of green coloration may provide indirect insights into determinacy as it reflects how primary growth is distributed across the season — more extended or bimodal green coloration patterns may indicate indeterminate growth. However, green coloration is a crude proxy and may confound photosynthetic activity with actual organogenesis or elongation, so its interpretation as a (in)determinacy measure should be treated with caution.

Future directions

We argue that incorporating plant determinacy into models of forest dynamics could greatly improve predictions of how climate change will affect ecosystem composition and productivity. The impacts of both climatic extremes and longer growing seasons on primary and secondary growth may depend strongly on the relative abundance of determinate versus indeterminate species within a forest community. Better understanding these two strategies and their response to a warmer and a drier climate will help to reveal the potential opportunities and critical limits of trees to adapt in a future climate, and may inform species or provenance choices in forest management.

One critical knowledge gap lies in understanding how different degrees of (in)determinacy help trees avoid or buffer environmental stress, while still maintaining the capacity to resume growth, repair tissues, replenish reserves, and compensate for earlier losses within the same season. Clarifying the trade-off between escaping unfavorable conditions and exploiting longer growing seasons is key to anticipating how forest communities will respond and reassemble over the coming decades.

Understanding how universally tissues within species are formed determinately or indeterminately could provide insight into the evolution of this trait, which could be aided by cross-species analyses. Both growth forms seem to occur across most species of the same family or clade and hence, appear to have evolved repeatedly in different groups (Figure **??**), making it difficult to speculate on which strategy is ancestral (but see Hariharan *et al.*, 2016) and how rapidly (in)determinacy can evolve. As data on which species are determinate or indeterminate accumulates (see Box 1), such analyses could potentially provide rapid insights into this, and also aid forecasting for unsampled species.

Moving forward, we need to characterize the plasticity of indeterminate growth across environmental gradients, species and populations with a particular focus on how much they are under genetic control. This will require integrative research combining genetics, physiology, ecology, and phenology. Moreover, we must assess how the prevalence of indeterminate growth affects forest-scale carbon dynamics, especially if (in)determinacy proves to be a continuous trait rather than a binary one.

Finally, we need to link the temporal dynamics of primary (apical and root) and secondary (cambial) meristems. Correlating annual tree rings with shoot increments could reveal such a common pattern, provided that the timing of inter-annual shoot segment (phytomer) production is accounted for, e.g. by distinguishing between phytomers that are preformed and those that are neoformed (Figure **??**). Revealing the patterns of when different meristems are active will likely contribute to a theoretical framework of temporal carbon allocation dynamics, and thus ultimately improve predictions on future forest carbon sequestration.

Acknowledgments

This text emerged from a grant of the Swiss National Foundation to F.B. (grant number P500PB_210943). We thank J. Ngo for designing figure 2.

Bibliography

Abramoff, R.Z. & Finzi, A.C. (2015). Are above- and below-ground phenology in sync? *New Phytologist*, 205, 1054–1061.

Arend, M., Hoch, G. & Kahmen, A. (2024). Stem growth phenology, not canopy greening, constrains deciduous tree growth. *Tree Physiology*, 44, tpad160.

Barthélémy, D. & Caraglio, Y. (2007). Plant Architecture: A Dynamic, Multilevel and Comprehensive Approach to Plant Form, Structure and Ontogeny. *Annals of Botany*, 99, 375–407.

Baumgarten, F., Gessler, A. & Vitasse, Y. (2023). No risk—no fun: Penalty and recovery from spring frost damage in deciduous temperate trees. *Functional Ecology*, 37, 648–663.

Baumgarten, F., Zohner, C.M., Gessler, A. & Vitasse, Y. (2021). Chilled to be forced: The best dose to wake up buds from winter dormancy. *New Phytologist*, 230, 1366–1377.

Böhlenius, H., Huang, T., Charbonnel-Campaa, L., Brunner, A.M., Jansson, S., Strauss, S.H. & Nilsson, O. (2006). CO/FT Regulatory Module Controls Timing of Flowering and Seasonal Growth Cessation in Trees. *Science*, 312, 1040–1043.

Boojh, R. & Ramakrishnan, P.S. (1982). Growth strategy of trees related to successional status I. Architecture and extension growth. *Forest Ecology and Management*, 4, 359–374.

Borchert, R. (1973). Simulation of Rhythmic Tree Growth under Constant Conditions. *Physiologia Plantarum*, 29, 173–180.

Borchert, R. (1976). The concept of juvenility in woody plants. *Acta Horticulturae*, pp. 21–36.

Brienen, R.J.W., Caldwell, L., Duchesne, L., Voelker, S., Barichivich, J., Baliva, M., Ceccantini, G., Di Filippo, A., Helama, S., Locosselli, G.M., Lopez, L., Piovesan, G., Schöngart, J., Villalba, R. & Gloor, E. (2020). Forest carbon sink neutralized by pervasive growth-lifespan trade-offs. *Nature Communications*, 11, 4241.

Buttò, V., Shishov, V., Tychkov, I., Popkova, M., He, M., Rossi, S., Deslauriers, A. & Morin, H. (2020). Comparing the Cell Dynamics of Tree-Ring Formation Observed in Microcores and as Predicted by the Vaganov–Shashkin Model. *Frontiers in Plant Science*, 11.

Caffarra, A. & Donnelly, A. (2011). The ecological significance of phenology in four different tree species: Effects of light and temperature on bud burst. *International Journal of Biometeorology*, 55, 711–721.

Campioli, M., Marchand, L.J., Zahnd, C., Zuccarini, P., McCormack, M.L., Landuyt, D., Lorer, E., Delpierre, N., Gričar, J. & Vitasse, Y. (2024). Environmental Sensitivity and Impact of Climate Change on leaf-, wood- and root Phenology for the Overstory and Understory of Temperate Deciduous Forests. *Current Forestry Reports*, 11, 1.

Chapin, F.S., Schulze, E. & Mooney, H.A. (1990). The Ecology and Economics of Storage in Plants. *Annual Review of Ecology and Systematics*, 21, 423–447.

Critchfield, W.B. (1971). Shoot Growth and Heterophylly in Acer. *Journal of the Arnold Arboretum*, 52, 240–266.

Damascos, M.A., Prado, C.H.B.A. & Ronquim, C.C. (2005). Bud Composition, Branching Patterns and Leaf Phenology in Cerrado Woody Species. *Annals of Botany*, 96, 1075–1084.

Delpierre, N., Vitasse, Y., Chuine, I., Guillemot, J., Bazot, S., Rutishauser, T. & Rathgeber, C.B.K. (2016). Temperate and boreal forest tree phenology: From organ-scale processes to terrestrial ecosystem models. *Annals of Forest Science*, 73, 5–25.

Ejsmond, M.J., Czarnołęski, M., Kapustka, F. & Kozłowski, J. (2010). How to Time Growth and Reproduction during the Vegetative Season: An Evolutionary Choice for Indeterminate Growers in Seasonal Environments. *The American Naturalist*, 175, 551–563.

Estiarte, M. & Peñuelas, J. (2015). Alteration of the phenology of leaf senescence and fall in winter deciduous species by climate change: Effects on nutrient proficiency. *Global Change Biology*, 21, 1005–1017.

Etzold, S., Sterck, F., Bose, A.K., Braun, S., Buchmann, N., Eugster, W., Gessler, A., Kahmen, A., Peters, R.L. & Vitasse, Y. (2021). Number of growth days and not length of the growth period determines radial stem growth of temperate trees. *Ecology Letters*.

Gärtner, H. & Farahat, E. (2021). Cambial Activity of Moringa peregrina (Forssk.) Fiori in Arid Environments. *Frontiers in Plant Science*, 12.

Girard, F., Vennetier, M., Ouarmim, S., Caraglio, Y. & Misson, L. (2011). Polycyclism, a fundamental tree growth process, decline with recent climate change: The example of Pinus halepensis Mill. in Mediterranean France. *Trees*, 25, 311–322.

Guédon, Y., Puntieri, J.G., Sabatier, S. & Barthélémy, D. (2006). Relative Extents of Preformation and Neoformation in Tree Shoots: Analysis by a Deconvolution Method. *Annals of Botany*, 98, 835–844.

Hallé, F., Oldeman, R.A.A. & Tomlinson, P.B. (1978). *Tropical Trees and Forests*. Springer, Berlin, Heidelberg.

Hao, Z., Hao, F., Singh, V.P. & Zhang, X. (2018). Changes in the severity of compound drought and hot extremes over global land areas. *Environmental Research Letters*, 13, 124022.

Hariharan, I.K., Wake, D.B. & Wake, M.H. (2016). Indeterminate growth: Could it represent the ancestral condition? *Cold Spring Harbor Perspectives in Biology*, 8, a019174.

Heide, O.M. (1993). Daylength and thermal time responses of budburst during dormancy release in some northern deciduous trees. *Physiologia Plantarum*, 88, 531–540.

Heuret, P., Meredieu, C., Coudurier, T., Courdier, F. & Barthélémy, D. (2006). Ontogenetic trends in the morphological features of main stem annual shoots of Pinus pinaster (Pinaceae). *American Journal of Botany*, 93, 1577–1587.

Hsiao, T.C. (1973). Plant responses to water stress. *Annual review of plant physiology*, 24, 519–570.

Hu, Z., Wang, H., Dai, J., Ge, Q. & Lin, S. (2022). Stronger Spring Phenological Advance in Future Warming Scenarios for Temperate Species With a Lower Chilling Sensitivity. *Frontiers in Plant Science*, 13.

Huijser, P. & Schmid, M. (2011). The control of developmental phase transitions in plants. *Development*, 138, 4117–4129.

Jin, S., Zhang, W., Shao, J., Wan, P., Cheng, S., Cai, S., Yan, G. & Li, A. (2022). Estimation of Larch Growth at the Stem, Crown, and Branch Levels Using Ground-Based LiDAR Point Cloud. *Journal of Remote Sensing*, 2022.

Journé, V., Szymkowiak, J., Foest, J., Hacket-Pain, A., Kelly, D. & Bogdziewicz, M. (2024). Summer solstice orchestrates the subcontinental-scale synchrony of mast seeding. *Nature Plants*, 10, 367–373.

Karkach, A.S. (2006). Trajectories and models of individual growth. *Demographic Research*, 15, 347–400.

Kaya, Z., Adams, W.T. & Campbell, R.K. (1994). Adaptive significance of intermittent shoot growth in Douglas-fir seedlings. *Tree Physiology*, 14, 1277–1289.

Keenan, T.F. & Richardson, A.D. (2015). The timing of autumn senescence is affected by the timing of spring phenology: Implications for predictive models. *Global Change Biology*, 21, 2634–2641.

Kikuzawa, K. (1983). Leaf survival of woody plants in deciduous broad-leaved forests. 1. Tall trees. *Canadian Journal of Botany*, 61, 2133–2139.

Körner, C. (2008). Winter crop growth at low temperature may hold the answer for alpine treeline formation. *Plant Ecology & Diversity*, 1, 3–11.

Kovaleski, A.P., Londo, J.P. & Finkelstein, K.D. (2019). X-ray phase contrast imaging of Vitis spp. buds shows freezing pattern and correlation between volume and cold hardiness. *Scientific Reports*, 9, 14949.

Kozlowski, T.T. (2012). *Seed Germination, Ontogeny, and Shoot Growth*. Elsevier.

Kozlowski, T.T. & Pallardy, S.G. (1997). *Growth Control in Woody Plants*. Elsevier.

Lang, G.A., Early, J.D., Martin, G.C. & Darnell, R.L. (1987). Endo-, Para-, and Ecodormancy: Physiological Terminology and Classification for Dormancy Research. *HortScience*, 22, 371–377.

Laube, J., Sparks, T.H., Estrella, N., Hofler, J., Ankerst, D.P. & Menzel, A. (2014). Chilling outweighs photoperiod in preventing precocious spring development. *Global Change Biology*, 20, 170–82.

Lechowicz, M.J. (1984). Why Do Temperate Deciduous Trees Leaf Out at Different Times? Adaptation and Ecology of Forest Communities. *The American Naturalist*.

Leites, L. & Benito Garzón, M. (2023). Forest tree species adaptation to climate across biomes: Building on the legacy of ecological genetics to anticipate responses to climate change. *Global Change Biology*, 29, 4711–4730.

Luo, T., Liu, X., Zhang, L., Li, X., Pan, Y. & Wright, I.J. (2018). Summer solstice marks a seasonal shift in temperature sensitivity of stem growth and nitrogen-use efficiency in cold-limited forests. *Agricultural and Forest Meteorology*, 248, 469–478.

Lupi, C., Morin, H., Deslauriers, A. & Rossi, S. (2010). Xylem phenology and wood production: Resolving the chicken-or-egg dilemma. *Plant, Cell & Environment*, 33, 1721–1730.

Lyford, W. & Wilson, B. (1966). Controlled growth of forest tree roots: Technique and application. *Harvard Forest Paper*.

Makoto, K., Wilson, S.D., Sato, T., Blume-Werry, G. & Cornelissen, J.H.C. (2020). Synchronous and asynchronous root and shoot phenology in temperate woody seedlings. *Oikos*, 129, 643–650.

Marchand, L.J., Gričar, J., Zuccarini, P., Dox, I., Mariën, B., Verlinden, M., Heinecke, T., Prislan, P., Marie, G. & Lange, H. (2025). No winter halt in below-ground wood growth of four angiosperm deciduous tree species. *Nature Ecology & Evolution*, pp. 1–9.

Marks, P.L. (1975). On the Relation between Extension Growth and Successional Status of Deciduous Trees of the Northeastern United States. *Bulletin of the Torrey Botanical Club*, 102, 172–177.

McDaniel, C.N. (1992). Induction and Determination: Developmental Concepts. *Flowering Newsletter*, pp. 3–6.

Meier, A.R., Saunders, M.R. & Michler, C.H. (2012). Epicormic buds in trees: A review of bud establishment, development and dormancy release. *Tree Physiology*, 32, 565–584.

Meng, F., Felton, A.J., Mao, J., Cong, N., Smith, W.K., Körner, C., Hu, Z., Hong, S., Knott, J., Yan, Y., Guo, B., Deng, Y., Leisz, S., Dorji, T., Wang, S. & Chen, A. (2024). Consistent time allocation fraction to vegetation green-up versus senescence across northern ecosystems despite recent climate change. *Science Advances*, 10, eadn2487.

Millet, J., Bouchard, A. & Édelin, C. (1999). Relationship between architecture and successional status of trees in the temperate deciduous forest. *Écoscience*, 6, 187–203.

Moore, E. (1909). The Study of Winter Buds with Reference to Their Growth and Leaf Content. *Bulletin of the Torrey Botanical Club*, 36, 117–145.

O'Sullivan, O.S., Heskel, M.A., Reich, P.B., Tjoelker, M.G., Weerasinghe, L.K., Penillard, A., Zhu, L., Egerton, J.J.G., Bloomfield, K.J., Creek, D., Bahar, N.H.A., Griffin, K.L., Hurry, V., Meir, P., Turnbull, M.H. & Atkin, O.K. (2017). Thermal limits of leaf metabolism across biomes. *Global Change Biology*, 23, 209–223.

Poethig, R.S. (2003). Phase Change and the Regulation of Developmental Timing in Plants. *Science*, 301, 334–336.

Pugnaire, F.I., Serrano, LUIS. & Pardos, JOSE. (1999). Constraints by water stress on plant growth. *Handbook of plant and crop stress*, 2, 271–283.

Radville, L., McCormack, M.L., Post, E. & Eissenstat, D.M. (2016). Root phenology in a changing climate. *Journal of Experimental Botany*, 67, 3617–3628.

Richardson, A.D., Hollinger, D.Y., Dail, D.B., Lee, J.T., Munger, J.W. & O'keefe, J. (2009). Influence of spring phenology on seasonal and annual carbon balance in two contrasting New England forests. *Tree Physiology*, 29, 321–331.

Roskilly, B. & Aitken, S. (2024). Weak Local Adaptation to Climate in Seedlings of a Deciduous Conifer Suggests Limited Benefits and Risks of Assisted Gene Flow. *Evolutionary Applications*, 17, e70001.

Rossi, S., Deslauriers, A., Anfodillo, T., Morin, H., Saracino, A., Motta, R. & Borghetti, M. (2006). Conifers in cold environments synchronize maximum growth rate of tree-ring formation with day length. *New Phytologist*, 170, 301–310.

Rossi, S., Deslauriers, A., Gricar, J., Seo, J.W., Rathgeber, C.B.K., Anfodillo, T., Morin, H., Levanic, T., Oven, P. & Jalkanen, R. (2008). Critical temperatures for xylogenesis in conifers of cold climates. *Global Ecology and Biogeography*, 17, 696–707.

Rudolph, T.D. (1964). Lammas Growth and Prolepsis in Jack Pine in the Lake States. *Forest Science*, 10, a0001–70.

Sakai, A. & Larcher, W. (1987). *Freezing Injuries in Plants*, Springer Berlin Heidelberg, Berlin, Heidelberg, vol. 62, pp. 39–58.

Schenker, G., Lenz, A., Körner, C. & Hoch, G. (2014). Physiological minimum temperatures for root growth in seven common European broad-leaved tree species. *Tree Physiology*, 34, 302–313.

Silvestro, R., Deslauriers, A., Prislan, P., Rademacher, T., Rezaie, N., Richardson, A.D., Vitasse, Y. & Rossi, S. (2025). From Roots to Leaves: Tree Growth Phenology in Forest Ecosystems. *Current Forestry Reports*, 11, 12.

Silvestro, R., Zeng, Q., Buttò, V., Sylvain, J.D., Drolet, G., Mencuccini, M., Thiffault, N., Yuan, S. & Rossi, S. (2023). A longer wood growing season does not lead to higher carbon sequestration. *Scientific Reports*, 13, 4059.

Soolanayakanahally, R.Y., Guy, R.D., Silim, S.N. & Song, M. (2013). Timing of photoperiodic competency causes phenological mismatch in balsam poplar (Populus balsamifera L.). *Plant, Cell & Environment*, 36, 116–127.

Soolanayakanahally, R.Y., Guy, R.D., Street, N.R., Robinson, K.M., Silim, S.N., Albrectsen, B.R. & Jansson, S. (2015). Comparative physiology of allopatric Populus species: Geographic clines in photosynthesis, height growth, and carbon isotope discrimination in common gardens. *Frontiers in plant science*, 6, 528.

Stearns, S.C. (1989). Trade-Offs in Life-History Evolution. *Functional Ecology*, 3, 259.

Stearns, S.C. (1998). *The Evolution Of Life Histories*. Oxford University Press.

Stevens, M.T. & Lindroth, R.L. (2005). Induced resistance in the indeterminate growth of aspen (Populus tremuloides). *Oecologia*, 145, 297–305.

Valdovinos-Ayala, J., Robles, C., Fickle, J.C., Pérez-de-Lis, G., Pratt, R.B. & Jacobsen, A.L. (2022). Seasonal patterns of increases in stem girth, vessel development, and hydraulic function in deciduous tree species. *Annals of Botany*, 130, 355–365.

Vitasse, Y., Baumgarten, F., Zohner, C.M., Rutishauser, T., Pietragalla, B., Gehrig, R., Dai, J., Wang, H., Aono, Y. & Sparks, T.H. (2022). The great acceleration of plant phenological shifts. *Nature Climate Change*, 12, 300–302.

Wang, Q., Liu, W., Leung, C.C., Tarté, D.A. & Gendron, J.M. (2024). Plants distinguish different photoperiods to independently control seasonal flowering and growth. *Science*, 383, eadg9196.

Zani, D., Crowther, T.W., Mo, L., Renner, S.S. & Zohner, C.M. (2020). Increased growing-season productivity drives earlier autumn leaf senescence in temperate trees. *Science*, 370, 1066–1071.

Zhang, Y., Wu, H. & Yang, W. (2019). Forests Growth Monitoring Based on Tree Canopy 3D Reconstruction Using UAV Aerial Photogrammetry. *Forests*, 10, 1052.

Zohner, C.M., Mirzagholi, L., Renner, S.S., Mo, L., Rebindaine, D., Bucher, R., Palouš, D., Vitasse, Y., Fu, Y.H., Stocker, B.D. & Crowther, T.W. (2023). Effect of climate warming on the timing of autumn leaf senescence reverses after the summer solstice. *Science*, 381, eadf5098.

Zohner, C.M., Mo, L., Pugh, T.A.M., Bastin, J.F. & Crowther, T.W. (2020a). Interactive climate factors restrict future increases in spring productivity of temperate and boreal trees. *Global Change Biology*, 26, 4042–4055.

Zohner, C.M., Mo, L., Renner, S.S., Svenning, J.C., Vitasse, Y., Benito, B.M., Ordonez, A., Baumgarten, F., Bastin, J.F. & Sebald, V. (2020b). Late-spring frost risk between 1959 and 2017 decreased in North America but increased in Europe and Asia. *Proceedings of the National Academy of Sciences*, 117, 12192–12200.

Zohner, C.M., Renner, S.S., Sebald, V. & Crowther, T.W. (2021). How changes in spring and autumn phenology translate into growth-experimental evidence of asymmetric effects. *Journal of Ecology*, 109, 2717–2728.